Started on	Friday, 29 November 2024, 2:15 PM
State	Finished
Completed on	Friday, 29 November 2024, 2:44 PM
Time taken	28 mins 51 secs
Marks	3.00/3.00
Grade	**3.00** out of 3.00 (**100**%)

Information

Information

This page contains all the problems for this test. The very last problem asks you to contact the person in charge of the exam and tell him or her the 4-digit key given in the problem text. In return you will be given a 5-digit signing code which you must give as the answer to the problem.

This problem does not count towards the final score, but **tests missing this code will not count towards the final grade**.

The following rules apply:

- Total time allowed: 30 minutes. The test will automatically close if time runs out.
- UiA's usual rules in regards to cheating on exams apply.

Question **1**

Correct

Mark 1.00 out of 1.00

Bob is using the RSA encryption scheme with public key equal to $(n, x) = (1541, 475)$.

a) Bob's RSA key is weak. Break his encryption and find his Bob's private key:

$y =$ 1183

Your last answer was interpreted as follows:

$$1183$$

It might help to know that the following is a list of the first prime numbers:

$$[2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79]$$

b) Alice uses Bob's public key to encrypt a message, which she then sends over a public channel.

The encrypted message is

$$M = 1227$$

Find the unencrypted (plain text) message

$m =$ 1232

Your last answer was interpreted as follows:

$$1232$$

The following calculator can be used for computing exponents $\mod (n)$:

Base:

Exponent:

Modulus:

Result:

Question **2**

Correct

Mark 1.00 out of 1.00

Given a noisy binary symmetric channel. The probability of making an error when transmitting a single bit is $p = 0.097$.

The bitstring $w =$**0000101** is transmitted.

In all of the answers below, all answers must be rounded off to the three decimals.

a) What is the probability of making no errors transmitting w?

0.490

Your last answer was interpreted as follows:

$$0.490$$

b) What is the probability of making at most 2 bit errors transmitting w?

0.976

Your last answer was interpreted as follows:

$$0.976$$

b) What is the probability of transmitting **0000101** and receiving $r =$ **0000000** ?

0.006

Your last answer was interpreted as follows:

$$0.006$$

Question **3**

Correct

Mark 1.00 out of 1.00

Let

$$E : (\mathbb{Z}/2)^{\times 3} \to (\mathbb{Z}/2)^{\times 5}$$

be the encoding function $E(w) = w \cdot G$, given by the generator matrix

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}.$$

a) Compute the minimum distance between any two codewords which are not equal.

l_{min}= 1

Your last answer was interpreted as follows:

$$1$$

b) Find a **non-zero** codeword c for the encoding E.

$c =$ 00100

Your last answer was interpreted as follows:

$$00100$$

Any codeword will do, as long as it is not the bitstring consisting only of zeros **000...0**.

c) Find a bitstring x in $(\mathbb{Z}/2)^{\times 5}$ which is **not** a codeword for the encoding E.

$x =$ 00001

Your last answer was interpreted as follows:

$$00001$$

Syntax guide: Give your answers to **b)** and **c)** as bitstrings. E.g. if you think the answer to a question is "1011", then you should enter **1011** .

Question **4**

Correct

Mark 0.00 out of 0.00

Signing code

Before closing the test you must answer this problem with a signing code given to you by the person in charge of the test.

Tests missing this signing code will be ignored and will not count towards the final score.

Key: 273

Signing code: 11728

Your last answer was interpreted as follows:

$$11728$$

◄ Test 3 (topics 7-9: Recurrence relations, Introduction to group theory, The multiplicative group of integers mod(n) and Lagrange's theorem)

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